

Package ‘fluorojip’

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Type Package

Title Analysis of Chlorophyll a Fluorescence Transient Parameters

Version 0.1.1

Description Computes chlorophyll a fluorescence transient parameters from fluorescence summary data, including minimum and maximum fluorescence, selected transient steps, and area. Provides standard photosynthetic performance indices and fluxes per reaction center and per cross section. Includes helpers to read supported exported CSV trace tables and validation workflows based on bundled example files, as well as visualization tools and an interactive Shiny interface.

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calc_fluorojip	<i>Compute chlorophyll a fluorescence OJIP / JIP-test parameters</i>
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Description

Computes chlorophyll a fluorescence OJIP / JIP-test parameters from fluorescence summary data.

Usage

```
calc_fluorojip(df)
```

Arguments

df	A data frame containing fluorescence summary columns. At minimum, provide Fo (fo or o), Fm (fm or p), J (j or fj), and I (i or fi). Reliable Mo, N, PI_abs, and RC-based fluxes require a K-step / 300 us-equivalent measurement supplied as k, fk, f300, or f300us.
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Details

The current implementation follows a summary-input workflow. Primary terms such as phi_Po, Vj, Vi, psi_Eo, phi_Eo, Mo, and PI_abs are computed explicitly from the supplied fluorescence summary values.

Mo is derived from the K-step / F300 region using the standard relation $4 * (F300 - Fo) / (Fm - Fo)$. If a K-step / 300 us-equivalent value is not available, the function does not attempt to guess Mo; instead, Mo, N, RC-based fluxes, and PI_abs are returned as NA with a warning.

The cross-section outputs ABS_CSm, TRo_CSm, ETo_CSm, and DIo_CSm are returned as operational package outputs for internal comparison. They are computed consistently from the supplied summary data, but they should be described carefully if compared against instrument-specific phenomenological flux conventions.

Value

A data frame containing the original input columns plus calculated FluorOJIP / JIP-test parameters.

calc_fluorojip_file	<i>Read a summary table and compute FluorOJIP parameters</i>
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Description

Read a summary table and compute FluorOJIP parameters

Usage

```
calc_fluorojip_file(file, sep = ";", dec = ".", ...)
```

Arguments

file	Path to the input summary table.
sep	Field separator used in the input file. Defaults to ";".
dec	Decimal mark used in the input file. Defaults to ".".
...	Additional arguments passed to <code>utils::read.csv()</code> .

Details

This is a convenience wrapper for summary-input workflows: `read.csv(file, ...) -> calc_fluorojip(df)`.

Value

A data frame containing the original input columns plus calculated FluorOJIP / JIP-test parameters.

calc_fluorojip_fluorpen	<i>Calculate FluorOJIP parameters from a FluorPen workbook</i>
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Description

Calculate FluorOJIP parameters from a FluorPen workbook

Usage

```
calc_fluorojip_fluorpen(file, sheet = 1)
```

Arguments

file	Path to the FluorPen .xlsx file.
sheet	Sheet name or numeric index. Defaults to the first sheet.

Details

This is a convenience wrapper for the typical FluorPen workflow: `read_fluorpen_xlsx(file) -> fluorpen_to_ojip(raw) -> calc_fluorojip(ojip)`.

Use this function when you want to import a supported FluorPen workbook and compute JIP-test parameters in one step.

Value

A data frame with FluorOJIP / JIP-test parameters calculated from the imported workbook.

`calc_fluorojip_handypea`

Calculate FluorOJIP parameters from a supported exported table

Description

Reads a supported exported trace table, derives an OJIP summary, and then computes FluorOJIP parameters.

Usage

```
calc_fluorojip_handypea(file)
```

Arguments

`file` Path to the supported Biolyzer-exported CSV trace table.

Details

Although the function name retains the historical handypea prefix for backward compatibility, the supported import workflow is based on Biolyzer-exported trace tables.

Typical workflow:

```
read_handypea_csv() -> handypea_to_ojip() -> calc_fluorojip().
```

Value

A data frame with FluorOJIP parameters returned by `calc_fluorojip()`.

`example_fluorojip`

Example OJIP summary data

Description

A dataset containing fluorescence summary values used to demonstrate the calculation of FluorOJIP / JIP-test parameters.

Usage

```
data(example_fluorojip)
```

Format

A data frame with columns for sample identification, treatment groups, and fluorescence summary values used in JIP-test calculations.

Details

The data are organized in the format expected by `calc_fluorojip()`, including sample identifiers, treatment groups, and summary fluorescence variables such as `fo`, `fm`, `j`, `i`, `area`, and, when available, `k`.

```
fluorojip_example_biolyzer_file
```

Get the bundled Biolyzer validation workbook path

Description

Returns the full path to the example Biolyzer workbook distributed with the package in `inst/extdata`. This file can be used in validation-oriented workflows that compare FluorOJIP outputs against vendor-calculated JIP-test parameters.

Usage

```
fluorojip_example_biolyzer_file()
```

Details

The bundled workbook is intended as a reproducible validation resource for supported Biolyzer-based workflows.

Value

A length-1 character vector containing the normalized full file path.

```
fluorpen_to_ojip
```

Convert FluorPen traces to an OJIP summary table

Description

Convert FluorPen traces to an OJIP summary table

Usage

```
fluorpen_to_ojip(x)
```

Arguments

`x` A list returned by `read_fluorpen_xlsx()`.

Details

This helper converts the FluorPen trace matrix into the fluorescence summary structure expected by the core calculation workflow, including the `O`, `K`, `J`, `I`, `P`, and area-related quantities extracted from the trace.

A typical workflow is: `read_fluorpen_xlsx(file) -> fluorpen_to_ojip(raw) -> calc_fluorojip(ojip)`.

Value

A data frame containing one row per sample with OJIP summary values ready for `calc_fluorojip()`.

handypea_to_ojip	<i>Convert a supported CSV trace export to an OJIP summary table</i>
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Description

Converts a supported exported trace table into an OJIP summary table ready for downstream FluorOJIP parameter calculation.

Usage

```
handypea_to_ojip(x)
```

Arguments

`x` A list object returned by `read_handypea_csv()`.

Details

Although the function name retains the historical handypea prefix for backward compatibility, the supported import workflow is based on Biolyzer-exported trace tables.

Time values are normalized internally to milliseconds so the function can target the standard OJIP steps even when exported trace times are recorded in seconds, milliseconds, or microseconds.

Value

A data frame containing `sample_id`, `t_fm`, `fo`, `k`, `fm`, `j`, `i`, `p`, and `area`.

normalized_jiptable	<i>Build a normalized JIP parameter table</i>
---------------------	---

Description

Creates a wide or long table of FluorOJIP / JIP-test parameters for each sample, with optional normalization for exploratory analysis, comparison across treatments, and export.

Usage

```
normalized_jiptable(
  df,
  params = NULL,
  sample_col = "sample_id",
  group_col = "treatment",
  normalize = c("none", "zscore", "minmax", "control_ratio", "control_then_zscore"),
  control_level = NULL,
  output = c("wide", "long"),
  digits = 6
)
```

Arguments

df	A data frame, typically the output of <code>calc_fluorojip()</code> .
params	Character vector of parameter names to include. If <code>NULL</code> , a default set of commonly used JIP-test parameters is selected.
sample_col	Name of the sample identifier column.
group_col	Name of the grouping column, typically a treatment column.
normalize	Normalization method. One of "none", "zscore", "minmax", "control_ratio", or "control_then_zscore".
control_level	Level of <code>group_col</code> to be used as the control when a control-based normalization method is requested.
output	Output format: "wide" or "long".
digits	Number of decimal places used to round normalized parameter columns.

Details

This function is intended for exploratory analysis and reporting workflows in which selected FluorOJIP / JIP-test parameters need to be compared across samples or treatments on a common scale.

Parameters such as `PI_abs`, `Mo`, and `RC`-based fluxes depend on the availability of a K-step / 300 us-equivalent input (for example, `k` or `f300us`) in the original summary data used by `calc_fluorojip()`.

Cross-section outputs such as `ABS_CSm`, `TRo_CSm`, `ETo_CSm`, and `DIO_CSm` are package outputs intended for operational comparison workflows and may not fully match every instrument-specific convention.

Value

A data frame in wide or long format containing the selected JIP-test parameters after the requested normalization step.

<code>plot_3d_fluorojip</code>	<i>Plot a 3D scatter plot of OJIP parameters</i>
--------------------------------	--

Description

Creates an exploratory 3D scatter plot for exactly three selected FluorOJIP / JIP-test parameters.

Usage

```
plot_3d_fluorojip(
  res,
  params = c("Fv_Fm", "PI_abs", "area"),
  group_col = "treatment",
  normalize = TRUE
)

plot_param_3d(
  res,
  params = c("Fv_Fm", "PI_abs", "area"),
  group_col = "treatment",
```

```

    normalize = TRUE
  )

plot_param_surface3d(
  res,
  params = c("Fv_Fm", "PI_abs", "area"),
  group_col = "treatment",
  normalize = TRUE
)

```

Arguments

res	Data frame with OJIP / JIP-test results.
params	Character vector of exactly 3 parameters to plot.
group_col	Name of the column with treatment or grouping labels.
normalize	Logical indicating whether selected parameters should be z-score normalized before plotting.

Details

plot_param_3d() and plot_param_surface3d() are aliases of plot_3d_fluorojip(). The name plot_param_surface3d() is retained for backward compatibility, although the function produces a 3D scatter plot rather than an interpolated surface.

Value

Invisibly returns the object produced by `scatterplot3d::scatterplot3d()`.

plot_heatmap_fluorojip

Plot a heatmap of OJIP parameters

Description

Creates a heatmap for selected FluorOJIP / JIP-test parameters across samples, with optional z-score normalization.

Usage

```

plot_heatmap_fluorojip(
  res,
  params,
  sample_col = "sample_id",
  group_col = "treatment",
  scale = "zscore",
  main = "Normalized JIP-test parameter heatmap"
)

plot_param_heatmap(
  res,
  params,

```



```

    sample_col = "sample_id",
    group_col = "treatment",
    scale = "zscore",
    main = "Normalized JIP-test parameter heatmap"
  )

```

Arguments

<code>res</code>	Data frame with OJIP / JIP-test results.
<code>params</code>	Character vector of parameters to plot.
<code>sample_col</code>	Name of the column with sample IDs.
<code>group_col</code>	Name of the column with treatment or grouping labels.
<code>scale</code>	Normalization method. Supported values are "zscore" and "none".
<code>main</code>	Title of the plot.

Value

Invisibly returns the object produced by `stats::heatmap()`.

<code>read_fluorpen_xlsx</code>	<i>Read a FluorPen Excel export</i>
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Description

Reads a FluorPen .xlsx workbook exported in a wide vendor layout, where columns represent measurements, the first column stores the OJIP time grid, and footer rows contain vendor-calculated JIP-test parameters.

Usage

```
read_fluorpen_xlsx(file, sheet = 1)
```

Arguments

<code>file</code>	Path to the FluorPen .xlsx file.
<code>sheet</code>	Sheet name or numeric index. Defaults to the first sheet.

Details

This function reads a supported FluorPen workbook and prepares its trace and footer summary data for downstream OJIP conversion, validation, and comparison workflows.

A typical workflow is: `read_fluorpen_xlsx(file) -> fluorpen_to_ojip(raw) -> calc_fluorojip(ojip)`.

Value

A list with metadata, the raw trace matrix, and the vendor summary:

- sample_id measurement identifiers taken from the header row
- measurement_time acquisition timestamps
- protocol_id protocol labels such as OJIP
- times_us raw time grid in microseconds
- times_ms raw time grid converted to milliseconds
- mat numeric matrix with one row per sample and one column per time
- summary_raw footer summary as imported from the workbook
- summary_numeric footer summary converted to numeric values where possible

read_handypea_csv	<i>Read a supported Biolyzer-exported CSV trace table</i>
-------------------	---

Description

Reads a CSV trace table exported by Biolyzer and prepares it for downstream OJIP summary extraction.

Usage

```
read_handypea_csv(file)
```

Arguments

file	Path to the supported Biolyzer-exported CSV trace table.
------	--

Details

Although the function name retains the historical handypea prefix for backward compatibility, this workflow is based on supported exported trace tables rather than direct parsing of proprietary raw instrument files.

The function locates the trace header, extracts the time values, reads the numeric trace block, removes trailing non-data rows, and returns the result in a format suitable for handypea_to_ojip().

Value

A list with three elements: times_s, the extracted time points; ids, the trace identifiers; and mat, a numeric matrix containing the fluorescence traces.

run_fluorojip_app	<i>Run the fluorojip Shiny application</i>
-------------------	--

Description

Launches the bundled Shiny interface for interactive FluorOJIP / JIP-test workflows, including data import, parameter calculation, validation, visualization, normalization, and export.

Usage

```
run_fluorojip_app(...)
```

Arguments

... Additional arguments passed to `shiny::runApp()`.

Details

The bundled Shiny application provides an interactive environment for working with fluorescence summary inputs and supported import workflows. Depending on the installed app version, available features may include OJIP curve inspection, parameter selection, normalized 2D plots, heatmaps, 3D plots, validation tabs, export tools, and built-in help content.

This function starts the application distributed with the package and passes any additional arguments directly to `shiny::runApp()`.

Value

This function is called for its side effect of launching the bundled Shiny application.

write_normalized_jiptable	<i>Write a normalized JIP parameter table</i>
---------------------------	---

Description

Exports a normalized FluorOJIP / JIP-test parameter table to disk for use in spreadsheet software or downstream statistical workflows.

Usage

```
write_normalized_jiptable(df, file, ...)
```

Arguments

df	A data frame to export, typically generated by <code>normalized_jiptable()</code> .
file	Path to the output file.
...	Additional arguments passed to <code>utils::write.table()</code> .

Details

The table is written with a semicolon field separator (`sep = ";"`) and decimal point (`dec = "."`). This behavior is intentional and may be convenient for spreadsheet workflows, but it differs from the default behavior of `write.csv()`.

Value

This function is called for its side effect of writing a file to disk. It invisibly returns the output path.

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